Submodule 36.04: Perceptron (In Depth) (Hinglish)

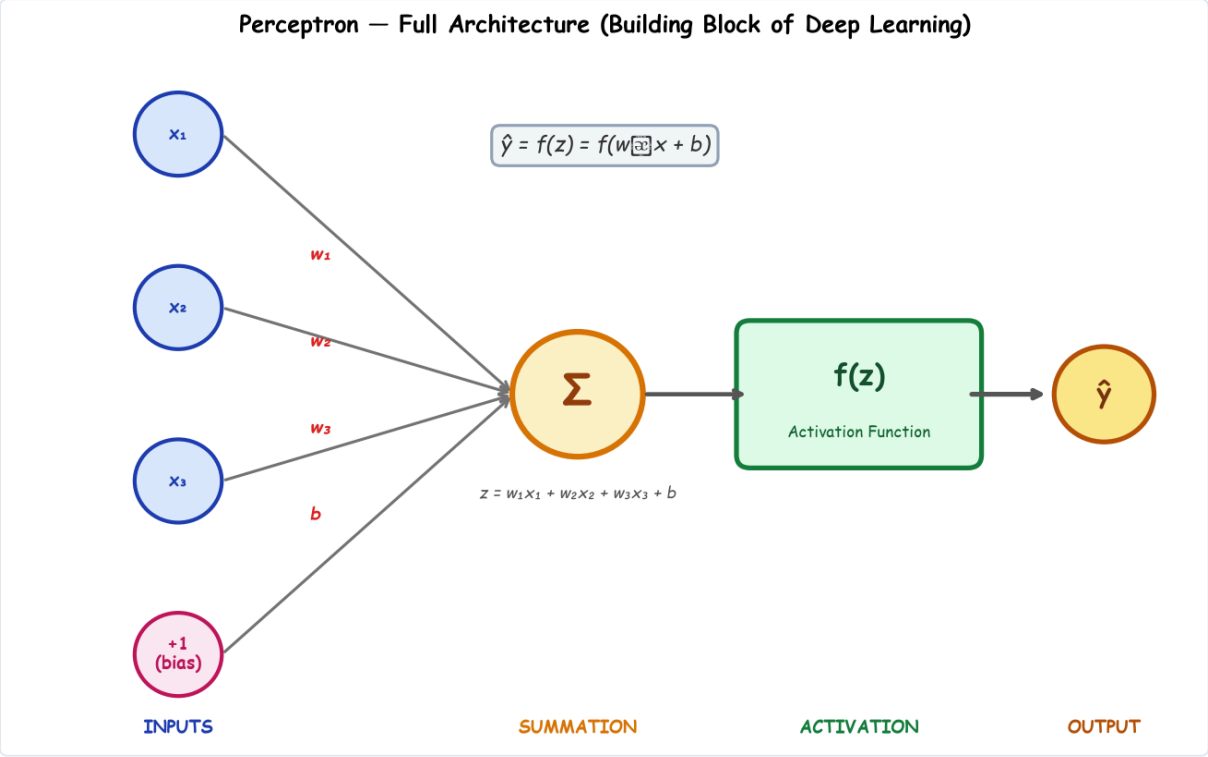
1. Perceptron ka Parichay

Perceptron Deep Learning aur Neural Networks ki sabse basic unit (single artificial neuron) hai. Ise 1958 mein **Frank Rosenblatt** ne invent kiya tha. Pura deep learning isi ek neuron ke mathematical formula par khada hai.

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2. Perceptron ka Mathematical Architecture

Perceptron inputs leta hai, unhe weights se multiply karta hai, bias jodta hai, aur activation function se pass karke output banata hai:



Formula ka Breakdown:

1. **Inputs ($x_1, x_2, \ldots$)**: Model ko milne wale raw features.
2. **Weights ($w_1, w_2, \ldots$)**: Har feature ki importance ko darshate hain. Jis feature ka weight zyada hoga, output par uska asar utna zyada hoga.
3. **Bias (b)**: Yeh ek constant value hoti hai jo sum me judti hai.
    - **Bias kyun zaroori hai?** Agar bias na ho ($b = 0$), toh decision boundary hamesha origin $(0, 0)$ se pass hogi. Bias boundary ko shift karne ki azaadi deta hai taaki agar

saare inputs zero bhi hon, tab bhi neuron activate ho sake.

4. **Weighted Sum (z)**:

$$z = \sum_{i=1}^{n} w_i x_i + b = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

Vector form mein:

$$z = \mathbf{w}^T \mathbf{x} + b$$

5. **Activation Function ($f(z)$)**:

$$\hat{y} = f(z)$$

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3. Activation Function ki Zaroorat Kyun Hai?

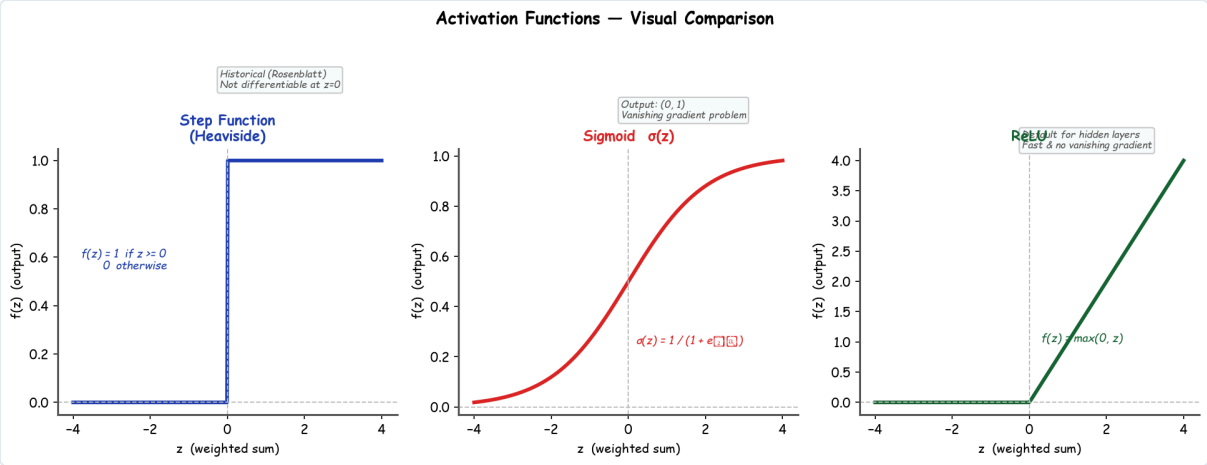
Agar hum activation function na lagayein, toh chahe aap 100 layers jod lo, mathematically wo ek single straight line (linear equation) hi rahegi:

$$W_2(W_1 x + b_1) + b_2 = W' x + b'$$

Real-world problems linear nahi hotin. **Activation functions non-linearity introduce karte hain**, jisse model complex curves aur shapes seekh pata hai.

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4. Teen Mukhya Activation Functions



1. **Step Function**:

$$f(z) = 1 extagar z \geq 0, extwarna 0$$

(Iska derivative zero hota hai, isliye modern gradient descent isme kaam nahi karta).

2. **Sigmoid Function**:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

(Output hamesha 0 *aur* 1 *ke beech hota hai, probability nikalne ke liye perfect).*

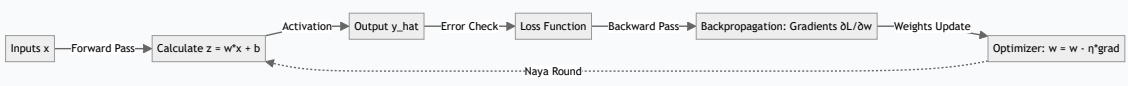
3. **ReLU (Rectified Linear Unit)**:

$$f(z) = \max(0, z)$$

(Negative values ko zero kar deta hai aur positive ko as-it-is chhod deta hai. Sabse fast aur popular).

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5. Neural Network Seekhta Kaise Hai? (Training Loop)



1. **Forward Pass**: Data inputs se start hokar formula ke zariye prediction $\hat{y}$ banata hai.
2. **Loss Function**: Check karta hai ki prediction sachai (y) se kitni door hai.

3. **Backpropagation**: Calculus ke chain rule se error ko peeche bhej kar gradients calculate karta hai.
4. **Optimizer**: Weights aur bias ko update karta hai taaki agle round me error kam ho:

$$w \leftarrow w - \eta \frac{\partial L}{\partial w}$$